PROJECT PROPOSAL FOR RESEARCH PROJECT

CST 498-6 / IIT 474-6

An Explainable AI Framework for Automated Identification and Education of *Devario* (Giant Danio) Species in Sri Lanka

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Declaration

We, hereby, declare that the research proposal titled "**An Explainable AI Framework for Automated Identification and Education of *Devario* (Giant Danio) Species in Sri Lanka**" is our original work and has been conducted solely for the purpose of fulfilling the partial fulfilment of the requirement of the course unit CST 498-6, Research Project for the degree of Computer Science and Technology.

We affirm that this work is free from any form of plagiarism and has not been submitted, in part or in full, for any other course, program, or qualification. All sources used in this proposal have been properly cited and acknowledged.

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List of Abbreviations

Artificial Intelligence	AI
Machine Learning	ML
Deep Learning	DL
Convolutional Neural Network	CNN
Ontology Web Language	OWL
Large Language Model	LLM
Retrieval-Augmented Generation	RAG
Computer Vision	CV
Explainable AI	XAI
International Union for Conservation of Nature	IUCN
Sri Lanka Department of Wildlife Conservation	DWC
Precision / Recall / F1-score	P/R/F1

1 Introduction

1.1 Project Background

Accurate species identification underpins ecology, biodiversity management, and conservation. Traditional morphology-based approaches are often unreliable for taxa with subtle or overlapping phenotypes (Teletchea 2009, Wäldchen & Mäder 2018). Sri Lanka hosts rich freshwater fish diversity, including several *Devario* species (giant danios)—many endemic and increasingly threatened (Sudasinghe et al. 2020, Pethiyagoda 1994). Misidentification also affects the ornamental fish trade, risking unlawful export of protected species and compounding transport-related mortality (Gunasekara 1996).

Recent advances in Machine Learning (ML) and Deep Learning (DL), especially Convolutional Neural Networks (CNNs), show strong performance for biological image classification, even under challenging conditions (Rahman et al. 2024, Rathi et al. 2017, Nair et al. 2018). Knowledge-based systems such as IPez demonstrate the value of combining morphometrics with ML (Guisande et al. 2010).

1.2 Problem Definition and Research Questions

The core problem is: *How can we create an intuitive, automated system that identifies and explains Devario species?* Specifically:

- How well can a CNN identify endemic freshwater fish from natural images, and what methods improve accuracy?
- How does combining image processing, an ontology, and ML improve speed and reliability?
- How can an ontology verify results and provide details such as habitat, threats, and significance?
- What challenges arise when using ontology-based explainable AI and LLM-powered querying at scale?

1.3 Significance and Rationale

An automated, explainable identification tool supports conservation practice, regulatory checks, and public education. It addresses the shortage of expert taxonomists (Hopkins & Freckleton 2002), reduces misidentification in trade, and promotes sustainable management. By fusing CNNs with an ontology and an LLM-driven Retrieval-Augmented Generation (RAG) chatbot, the system can both recognise species and explain the decision in human-friendly terms.

1.4 Aim and Objectives

Aim: Develop an intelligent, explainable AI system that integrates deep learning-based image classification with ontology-driven knowledge representation and an LLM-powered chatbot to accurately identify and explain six Sri Lankan and regional *Devario* species.

Objectives:

1. Build and train a CNN to classify target species (*D. acrostomus*, *D. acuticephala*, *D. salmonatus*, *D. strigillifer*, *D. yuensis*, *D. apogon*) from real-world images.
2. Construct an OWL ontology encoding taxonomy, morphology, habitat, and conservation status; expose SPARQL queries.
3. Implement a RAG chatbot that retrieves ontology facts and generates accurate, contextual responses.
4. Evaluate the system via accuracy, precision, recall, F1-score (CNN) and task-based usability/consistency checks (chatbot).

2 Project Description

2.1 Scope and Target Species

The project focuses on six *Devario* species relevant to Sri Lanka and the region: *D. acrostomus*, *D. acuticephala*, *D. salmonatus*, *D. strigillifer*, *D. yuensis*, and *D. apogon*. These giant danios exhibit diagnostic stripe/bar patterns and body proportions that a CNN can learn to discriminate.

2.2 System Overview

Users upload a fish image (e.g., smartphone photo). A CNN predicts the species label. The system then consults an ontology-backed knowledge base for structured facts (morphology, habitat, conservation status). A RAG chatbot combines those facts with a Large Language Model (LLM) to answer user questions (e.g., distribution, threats) in natural language while remaining grounded in the ontology.

2.3 Components

1. **Image Classifier (CNN):** Trained with a curated dataset and augmentations (rotation, flip, lighting) to handle variability.
2. **Ontology (OWL):** Encodes species traits, habitats, and IUCN status; supports SPARQL queries for verification and explanations.
3. **RAG Chatbot:** Retrieves ontology snippets and generates concise, contextual responses for education and decision support.
4. **Web/API Layer:** Handles uploads, runs inference, serves explanations, and logs feedback for iterative improvement.

2.4 Expected Outcomes

A working prototype for identification and explanation; a structured fish ontology; dashboards or simple reports visualising accuracy and common failure modes; and guidance for conservation and trade verification use cases.

3 Methodology

3.1 Approach

A design-and-development study integrating quantitative image analysis with a knowledge-based interface. We iterate through design, implementation, pilot testing, and evaluation.

3.2 Research Design

CNN Image Classification: Train (e.g., ResNet/EfficientNet) on labelled images; apply data augmentation and cross-validation.
Ontology Knowledge Base: Develop an OWL ontology capturing taxonomy, diagnostic morphology (e.g., stripe counts), habitat, and conservation status.
RAG Chatbot: Use ontology retrieval to ground an LLM's responses; enforce answer checks against retrieved triples for factual consistency.

3.3 Data Collection

Images: Diverse, labelled photos per species from authoritative or licensed sources and collaborators; preprocessing: resize, normalise, augment.
Texts: Taxonomic descriptions, FishBase/IUCN summaries, and peer-reviewed literature to populate ontology facts (converted into RDF triples).

3.4 Sampling Strategy

Purposive sampling to ensure coverage of pose, lighting, background, and intra-species variation. Target at least 100 distinct images per species. Prioritise peer-reviewed or recognised databases for textual facts.

3.5 Data Analysis

CNN: Optimise hyperparameters; report accuracy, precision, recall, F1-score. Use Grad-CAM or similar to visualise salient regions for explainability (Rathi et al. 2017, Nair et al. 2018).
Ontology + RAG: Query via SPARQL; ground LLM prompts with retrieved facts; verify outputs against the ontology to reduce hallucinations.

3.6 Ethical Considerations

Use publicly available/licensed data with citations; no live specimen handling. Communicate that identifications are probabilistic and should be expert-confirmed for high-stakes decisions.

3.7 Validity and Reliability

Use expert-verified labels where possible; hold-out validation; cross-validation; augmentation for robustness. Regular expert feedback loops; ontology facts sourced from recognised authorities.

3.8 Limitations

Scope limited to six *Devario* species; extreme image conditions may degrade accuracy; secondary-source errors can propagate; LLM outputs require grounding and verification.



Figure 1: Illustrative system architecture and data flow (placeholder).

4 Resources Needed

4.1 Software

- **Python** (TensorFlow/Keras or PyTorch) for CNN training; **OpenCV** for preprocessing.
- **Protégé** for ontology modelling (OWL) and **SPARQL** querying.
- **Grad-CAM/SHAP** for explainability; **FAISS/**vector store for retrieval in RAG.
- **Backend & API:** FastAPI/Flask or Node.js; simple web UI for uploads and results.

4.2 Hardware

- GPU-enabled workstation/cloud (e.g., Google Colab) for model training.
- Edge device option (e.g., Jetson Nano) for on-site inference (optional).

4.3 Storage

- Cloud/object storage for images, models, and ontology files; local SSD for fast training I/O.

4.4 People and Permissions

- Four student developers; one supervisor; optional domain expert (ichthyology).
- Ensure data licensing and ethical approval where applicable.

5 Individual Contribution and Work Plan

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6 Project Timeline (10 Months)

Placeholder timeline—add your months and tasks when finalised (kept empty per supervisor instruction).

- Month 1: *(fill later)*
- Months 2–3: *(fill later)*
- Months 4–5: *(fill later)*
- Month 6: *(fill later)*
- Months 7–8: *(fill later)*
- Month 9: *(fill later)*
- Month 10: *(fill later)*



Figure 2: Planned Gantt chart (placeholder).

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